

MINNEAPOLIS CRYSTAL, MN, USA

WMO: 726575

Lat: 45.062N

Lon: 93.351W

Elev: 861

StdP: 14.24

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 94960

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.7	-6.1	-18.3	2.1	-8.5	-15.1	2.5	-5.3	23.5	15.4	20.7	15.4	7.0	300	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	90.4	73.3	87.9	71.9	84.4	69.8	76.6	86.5	74.5	83.9	72.5	81.7	11.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	127.1	83.0	71.6	120.7	81.2	69.7	112.7	79.0	40.6	86.3	38.6	83.9	36.7	81.6	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	19.0	17.2	-16.9	95.8	5.0	3.1	-20.5	98.0	-23.5	99.8	-26.3	101.6	-30.0	103.8
			-17.3	79.9	4.9	2.5	-20.8	81.8	-23.7	83.2	-26.4	84.7	-30.0	86.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.3	16.2	19.8	32.4	46.5	58.4	68.1	73.7	70.7	63.1	48.7	34.9	21.4
	DBStd	22.25	12.45	12.04	12.26	10.13	8.61	6.85	5.62	5.52	8.52	9.33	10.31	11.70
	HDD50	4151	1049	846	561	181	21	0	0	0	6	138	463	886
	HDD65	7555	1514	1266	1013	559	241	45	4	12	135	512	903	1351
	CDD50	2800	0	0	14	76	282	543	734	642	400	99	10	0
	CDD65	730	0	0	1	3	37	139	274	189	79	8	0	0
	CDH74	6676	0	0	5	52	425	1237	2635	1583	672	67	0	0
	CDH80	2123	0	0	0	11	137	375	939	460	189	12	0	0
	WSAvg	8.0	8.2	8.3	8.5	9.1	8.4	7.7	7.2	6.8	7.7	8.1	8.3	8.0
	PrecAvg	32.0	0.9	0.8	1.7	3.0	4.1	4.8	4.6	4.3	3.3	2.8	1.8	1.1
Precipitation	PrecMax	42.0	2.0	1.6	3.7	7.8	9.1	8.9	10.0	7.5	8.5	6.4	5.4	2.4
	PrecMin	23.3	0.1	0.2	0.4	0.9	0.2	1.8	1.1	0.4	0.4	0.6	0.1	0.3
	PrecStd	4.3	0.5	0.4	0.8	1.5	2.0	1.8	2.1	1.9	2.1	1.6	1.4	0.6
	PrecStd	4.3	0.5	0.4	0.8	1.5	2.0	1.8	2.1	1.9	2.1	1.6	1.4	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.1	54.4	70.9	81.0	90.5	92.5	95.9	92.0	89.9	81.5	67.1	50.3
		MCWB	38.3	46.4	57.4	62.1	68.7	73.5	76.4	73.8	71.7	65.6	55.3	46.2
	2%	DB	39.0	45.3	62.5	72.7	83.6	88.0	90.8	88.3	84.9	74.4	61.1	44.6
		MCWB	34.9	40.1	53.0	55.8	65.1	70.8	75.0	72.7	70.1	61.3	51.7	39.4
	5%	DB	36.0	40.5	56.1	68.4	79.0	84.0	88.2	84.4	81.4	69.2	54.9	39.3
		MCWB	33.2	36.3	48.1	53.3	62.6	68.9	73.1	70.3	67.7	57.6	47.2	35.8
	10%	DB	33.5	36.6	49.9	63.8	73.4	81.2	84.9	81.9	77.5	63.7	50.4	36.5
		MCWB	31.2	33.3	43.6	50.6	59.3	66.7	71.1	68.8	65.8	55.2	44.0	33.7
	0.4%	WB	38.7	46.5	60.5	63.4	72.8	76.7	80.3	78.0	75.1	68.7	58.3	47.4
		MCDB	44.1	53.5	67.3	76.9	85.8	87.8	89.6	88.0	85.5	75.9	64.4	50.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.4	40.6	54.1	58.7	67.6	73.6	77.5	75.2	72.3	63.0	53.0	39.7
		MCDB	38.2	45.0	60.7	69.1	79.1	83.8	87.8	83.8	81.5	69.8	59.7	43.1
	5%	WB	33.1	36.7	49.1	55.0	64.9	71.1	75.3	73.1	69.8	59.5	47.8	36.4
		MCDB	35.6	40.3	56.2	65.6	74.9	81.0	84.6	81.9	77.8	67.0	54.1	39.5
	10%	WB	30.7	33.4	43.7	51.8	62.0	68.7	73.2	71.1	67.5	55.7	44.4	33.9
		MCDB	32.8	36.4	49.3	62.2	71.5	77.8	82.6	79.2	76.1	63.0	49.9	36.4
Mean Daily Temperature Range	5% DB	MDBR	14.7	15.9	16.9	19.5	20.2	18.9	19.2	18.9	19.6	18.0	15.2	13.3
		MCDBR	16.4	18.0	24.1	27.1	27.1	22.5	21.3	21.4	23.2	25.9	22.7	17.2
		MCWBR	13.6	13.6	15.4	14.8	12.7	10.2	9.5	9.4	11.2	14.7	15.0	13.7
	5% WB	MCDBR	15.5	17.1	22.6	23.8	23.2	19.4	19.0	18.5	19.4	20.9	21.1	15.6
		MCWBR	13.1	13.2	15.2	14.5	12.7	10.5	10.1	10.2	11.0	13.9	15.0	12.9
Clear-Sky Solar Irradiance	taub		0.270	0.283	0.320	0.378	0.409	0.400	0.399	0.406	0.377	0.322	0.292	0.269
	taud		2.412	2.363	2.381	2.289	2.254	2.328	2.354	2.334	2.378	2.527	2.512	2.435
	Ebn at Noon		272	288	290	279	273	276	274	268	267	271	262	260
	Edh at Noon		23	29	33	39	42	39	38	37	33	25	21	20
All-Sky Solar Radiation	RadAvg		546	891	1208	1460	1690	1881	1994	1663	1296	820	522	382
	RadStd		64	88	110	146	138	147	72	121	91	109	49	44

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page